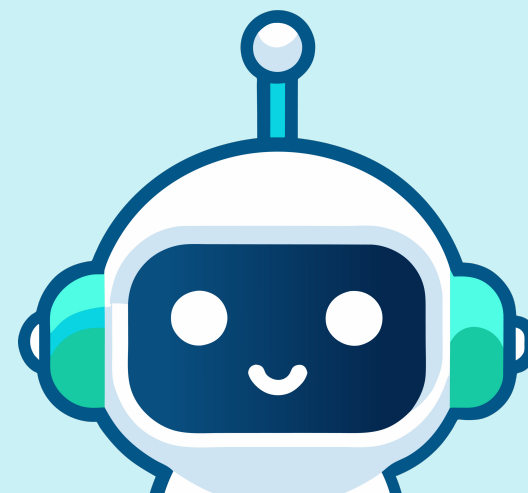


# NanoSense

Affordable, Portable, Reliable

Adriana, Audrey, Adam, Aarushi, Ishani, Juhi,  
Developed by A.A.A.A.I.J.R.H.

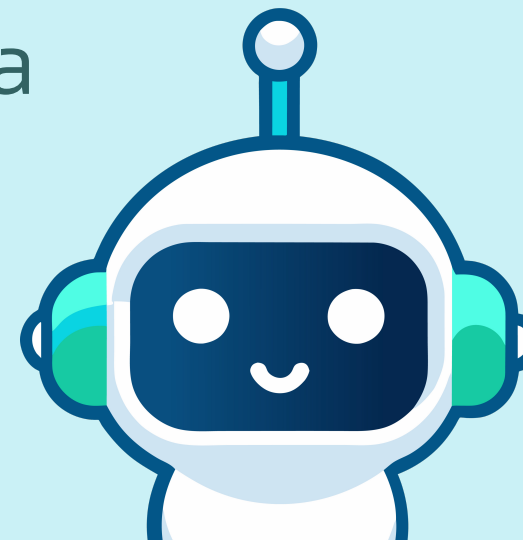


# FOCUS & PROBLEM

[Challenge Track & UN SDG #3]

**Health & Well-being: How can we have quality healthcare and well-being for all residents?**

- It's hard for many people to access quick and accurate medical diagnoses
- Poor-quality care is responsible for up to 15% of deaths in low- and middle-income countries
- Around the world, roughly 1/3 of total deaths are caused by preventable diseases
- Approximately 1 in 20 adults experience a diagnostic error each year





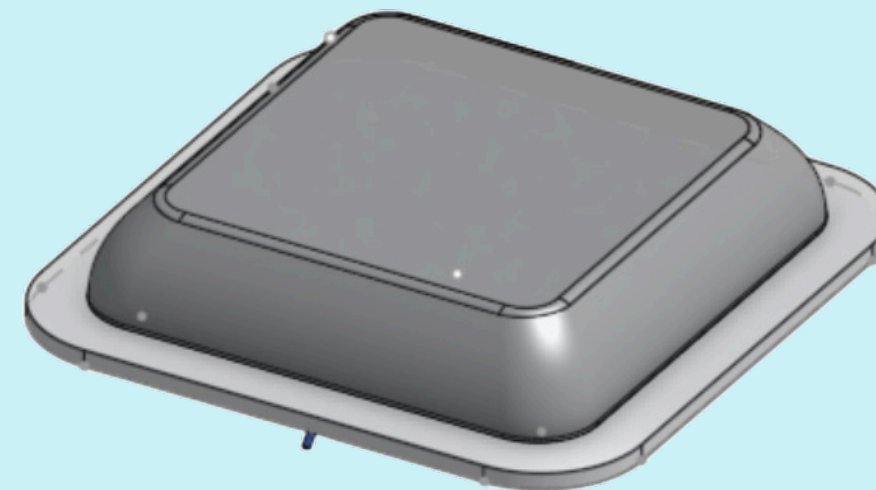
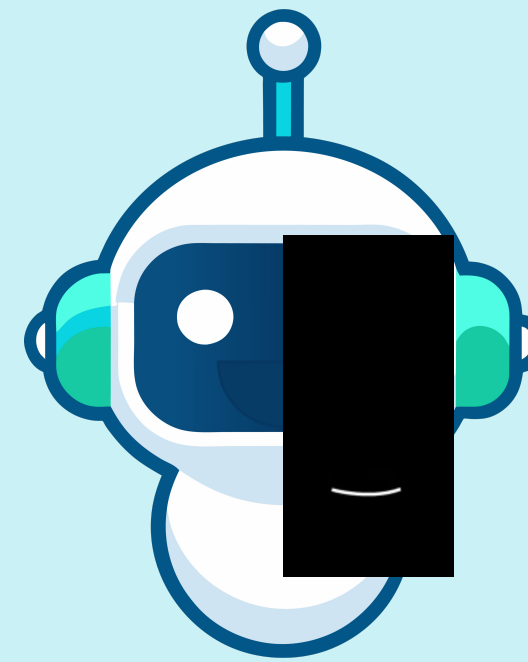
# OUR SOLUTION

- A biannual "check-up" provided to all residents
- For the purpose of early disease detection and therefore prevention
- Prioritizes accessibility and user convenience
  - Meant to be used in your own home
  - Does not require an appointment or any supervision from medical professionals
- Lowers the workload of understaffed facilities
  - Less staff doing easily automated "busywork" (namely patient data collection + analysis)



# THE PHYSICAL PRODUCT

- Nanobot which travels through the bloodstream
  - Installed with basic AI for navigation
  - Collects data similar to a blood sample test
  - Photographs abnormalities (ex: atherosclerosis)
- Injection device placed on the upper arm
  - Inspired by real-life technology, omnipods, which delivers insulin to diabetic patients
  - Travels through the bloodstream whilst collecting data from your blood samples



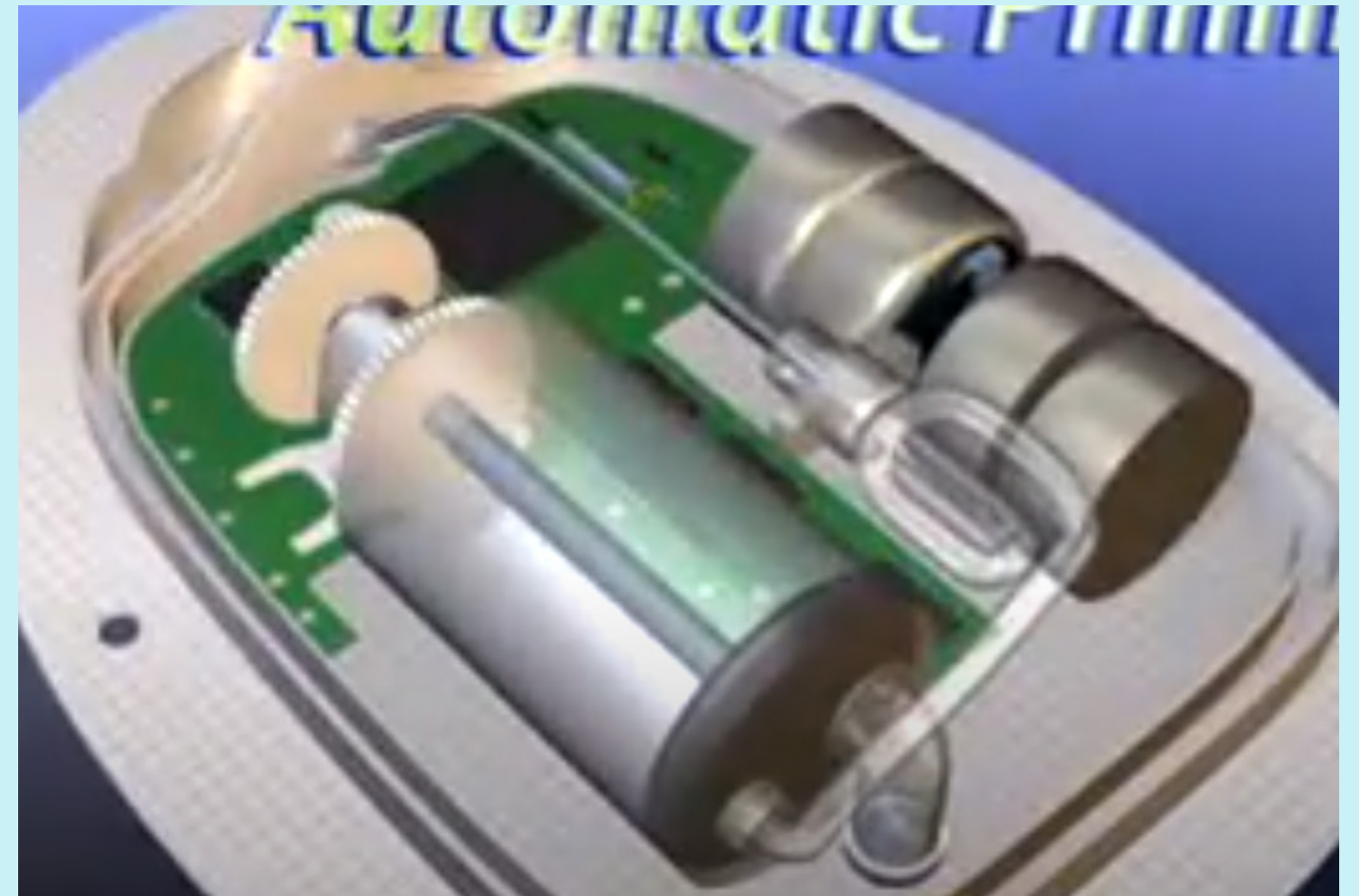
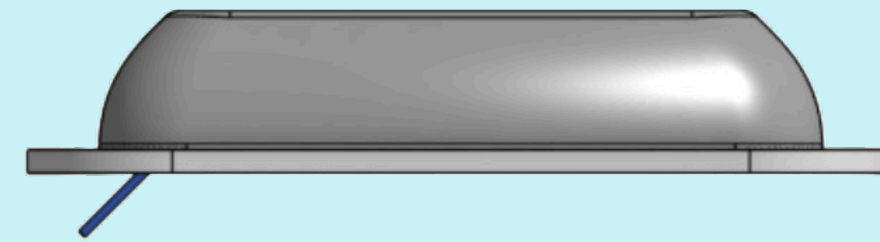


# PRODUCTION DETAILS

- Technology is nontoxic to the human body
  - Small enough in scale to get filtered out by the kidneys (expelled as waste)

## Materials:

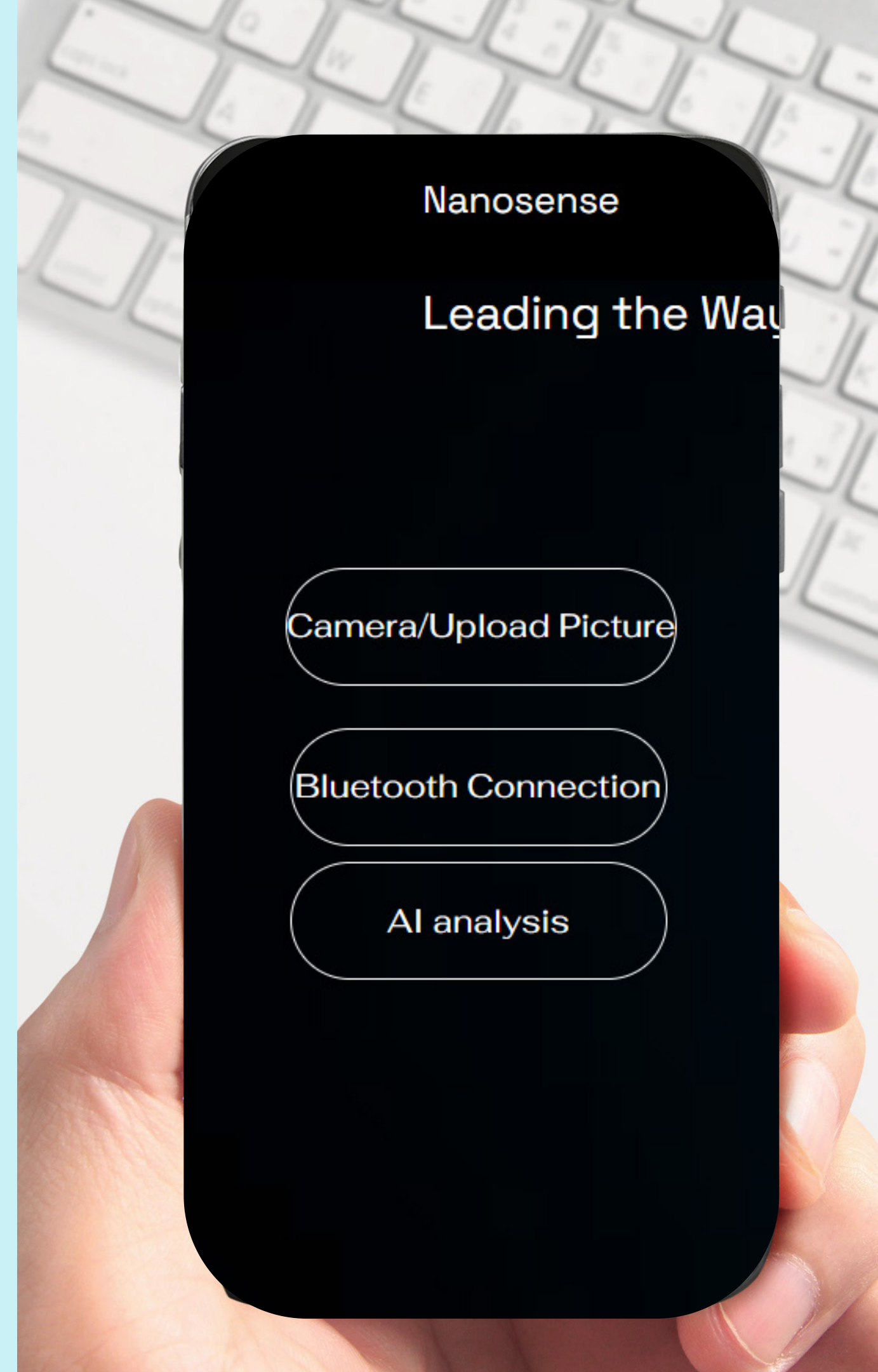
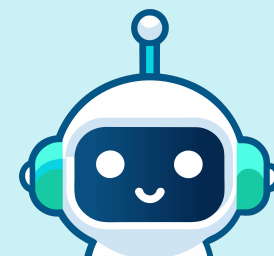
- Stainless Steel Needle
- Nanobot container (replacable)
- Electronics: Batteries, PCBs, et.
- Filtering tube
- Real-world platinum alloy, CoCr, which is utilized in pacemakers



# APP FEATURES

- Embedded with the latest AI advancements to analyze nanobot data for possible health risk
- Gives personalized lifestyle recommendations
- Automatically contacts your doctor with any concerns

[Link to website demo](#)





AI Welcome to At Home Blood Sample Analyzer! Are you ready to get started?



i see an irregular white cell

AI  **References used**

If you notice an irregular white cell in your blood sample, it is crucial to understand the context of the white blood cell (WBC) differential percentages to determine the stage of illness. Here's how we can proceed:

## Step 1: Analyze the Blood Sample

To accurately assess the stage of illness, we would need to know the specific counts or percentages of different types of white blood cells in your sample. However, here's a general guideline:

- **Neutrophils:** An increase generally indicates an acute bacterial infection.
- **Lymphocytes:** A decrease might suggest a more severe infection or progression towards sepsis.

# APP INTERFACE





# CODE BEHIND IT

```
def detect_anomaly(image):
```

```
    """
```

```
    Detect anomalies in the given image by analyzing red color concentrations.
```

```
    This function simulates anomaly detection by flagging regions with significant
    red intensity—which may indicate inflammation or abnormal tissue growth.
    It converts the image to HSV color space, applies color masks for red,
    and calculates the proportion of red pixels.
```

```
    Parameters:
```

```
    image (numpy.ndarray): The input image from the nano bot's camera.
```

```
    Returns:
```

```
    (bool, str): A tuple where the first element is True if an anomaly is detected,
    and the second element provides details about the detection.
```

```
    """
```

```
    # Convert image to HSV for better color segmentation.
```

```
    hsv_image = cv2.cvtColor(image, cv2.COLOR_BGR2HSV)
```

```
    # Define HSV ranges for red (covering the lower and upper parts of the hue range).
```

```
    lower_red1 = np.array([0, 70, 50])
```

```
    upper_red1 = np.array([10, 255, 255])
```

```
    lower_red2 = np.array([170, 70, 50])
```

```
    upper_red2 = np.array([180, 255, 255])
```

```
    # Create masks for red regions.
```

```
    mask1 = cv2.inRange(hsv_image, lower_red1, upper_red1)
```

```
    mask2 = cv2.inRange(hsv_image, lower_red2, upper_red2)
```

```
    red_mask = cv2.bitwise_or(mask1, mask2)
```

```
    # Calculate the ratio of red pixels to the total number of pixels.
```

```
    red_pixels = np.sum(red_mask > 0)
```

```
    total_pixels = image.shape[0] * image.shape[1]
```

```
    red_ratio = red_pixels / total_pixels
```

```
    # Flag an anomaly if more than 5% of the image is red (this threshold is
    adjustable).
```

```
    if red_ratio > 0.05:
```

```
        message = (f"Anomaly detected: {red_ratio:.2%} of the image shows
        red areas. "
```

```
                "This may indicate inflammation or abnormal tissue.")
```

```
        return True, message
```

```
    else:
```

```
        return False, "No significant anomalies detected."
```

```
import cv2
```

```
import numpy as np
```

```
def load_image(image_path):
```

```
    """
```

```
    Load an image from the specified file path.
```

```
    This function simulates capturing an image from a nano bot's camera.
```

```
    Parameters:
```

```
    image_path (str): The path to the image file.
```

```
    Returns:
```

```
    numpy.ndarray: The loaded image.
```

```
    """
```

```
    image = cv2.imread(image_path)
```

```
    if image is None:
```

```
        raise ValueError("Could not load image. Check the path provided.")
```

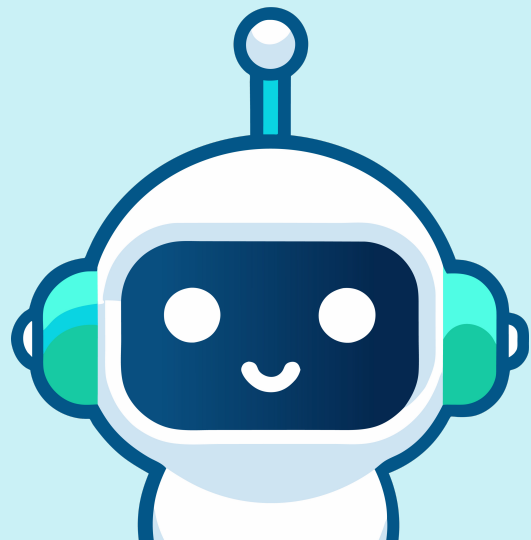
```
    return image
```





# Q&A

*Are there any questions?*



# THANK YOU

*Are there any questions?*

